Technology Stack

Date	30 September 2026
Team ID	6ab3ed08a238dc999eb3fda5
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	React.js with HTML5, CSS3 and Tailwind CSS	Component-based UI that is reusable and fast to update; responsive design works on mobile and desktop; accessible components support WCAG 2.1 AA.
2	**Backend / Server-Side**	Node.js with Express.js (REST APIs)	Fast, non-blocking processing suits many concurrent learners; JavaScript across frontend and backend simplifies development; large library ecosystem.
3	**Database / Data Storage**	PostgreSQL Amazon S3 (files)	Strong relational mapping for users, courses, enrollments and credentials, with reliable transactions for payments and certificates. S3 stores course material and certificate PDFs.
4	**Cloud / Hosting / Deployment**	AWS (EC2 / ECS, RDS) Docker	Easy scalability to meet the 10,000 concurrent user target and high availability; Docker gives identical environments from development to production.

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5	**Version Control & CI/CD**	GitHub GitHub Actions	Seamless team collaboration through branches and pull requests; automated testing and deployment on every code push.
6	**Third-Party APIs / Other Tools**	Razorpay Google OAuth 2.0 LinkedIn Share API Twilio (SMS) / Amazon SES (email)	Razorpay enables secure course payments; Google OAuth gives one-click sign-in; LinkedIn lets learners share verified credentials; Twilio and SES send OTPs, alerts and notifications.